

Disseny d'interfícies web

RA4. Elements multimèdia

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Actualment CSS3 no només és una potent eina per maquetar una pàgina web mitjançant tècniques com flex i grid.

També ens permet crear una gran varietat d'animacions sense la necessitat d'utilitzar javascript.

Permet que els element es puguin moure, girar, escalar o inclinar tant a nivell de l'eix de les X i Y (2 dimensions)

Function	Description
<code>matrix(<i>n,n,n,n,n,n</i>)</code>	Defines a 2D transformation, using a matrix of six values
<code>translate(<i>x,y</i>)</code>	Defines a 2D translation, moving the element along the X- and the Y-axis
<code>translateX(<i>n</i>)</code>	Defines a 2D translation, moving the element along the X-axis
<code>translateY(<i>n</i>)</code>	Defines a 2D translation, moving the element along the Y-axis
<code>scale(<i>x,y</i>)</code>	Defines a 2D scale transformation, changing the elements width and height
<code>scaleX(<i>n</i>)</code>	Defines a 2D scale transformation, changing the element's width
<code>scaleY(<i>n</i>)</code>	Defines a 2D scale transformation, changing the element's height
<code>rotate(<i>angle</i>)</code>	Defines a 2D rotation, the angle is specified in the parameter
<code>skew(<i>x-angle,y-angle</i>)</code>	Defines a 2D skew transformation along the X- and the Y-axis
<code>skewX(<i>angle</i>)</code>	Defines a 2D skew transformation along the X-axis
<code>skewY(<i>angle</i>)</code>	Defines a 2D skew transformation along the Y-axis

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https://www.w3schools.com/css/css3_2dtransforms.asp

Transformacions 3d

També permet modificar l'eix de les Z (3 dimensions)

Function	Description
<code>matrix3d</code> <code>(n,n,n,n,n,n,n,n,n,n,n,n,n,n,n,n)</code>	Defines a 3D transformation, using a 4x4 matrix of 16 values
<code>translate3d(x,y,z)</code>	Defines a 3D translation
<code>translateX(x)</code>	Defines a 3D translation, using only the value for the X-axis
<code>translateY(y)</code>	Defines a 3D translation, using only the value for the Y-axis
<code>translateZ(z)</code>	Defines a 3D translation, using only the value for the Z-axis
<code>scale3d(x,y,z)</code>	Defines a 3D scale transformation
<code>scaleX(x)</code>	Defines a 3D scale transformation by giving a value for the X-axis
<code>scaleY(y)</code>	Defines a 3D scale transformation by giving a value for the Y-axis
<code>scaleZ(z)</code>	Defines a 3D scale transformation by giving a value for the Z-axis
<code>rotate3d(x,y,z,angle)</code>	Defines a 3D rotation
<code>rotateX(angle)</code>	Defines a 3D rotation along the X-axis
<code>rotateY(angle)</code>	Defines a 3D rotation along the Y-axis
<code>rotateZ(angle)</code>	Defines a 3D rotation along the Z-axis
<code>perspective(n)</code>	Defines a perspective view for a 3D transformed element

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https://www.w3schools.com/css/css3_3dtransforms.asp

La propietat CSS overflow controla què passa amb el contingut que és massa gran per cabre dins d'una àrea.

overflow: visible

You can use the overflow property when you want to have better control of the layout. The overflow property specifies what happens if content overflows an element's box.

overflow: hidden

You can use the overflow property when you want to have better control of the layout. The overflow property specifies what

overflow: scroll

the layout. The overflow property specifies what happens if content overflows an element's box.

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https://www.w3schools.com/css/css_overflow.asp

shadow : Podem utilitzar aquesta propietat pels textos i els divs.

- url: https://www.w3schools.com/css/css3_shadows.asp

filter: Podem utilitzar aquesta propietat per les imatges, textos i divs.

- url: https://www.w3schools.com/cssref/css3_pr_filter.php

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https://www.w3schools.com/css/css3_shadows.asp

https://www.w3schools.com/cssref/css3_pr_filter.php

Quan un element fa un **canvi en una de les seves propietats**, aquesta tècnica **permet que el canvi en comptes de fer-se de forma instantània es faci a través d'un efecte de transició**.

Per fer aquesta transició CSS necessita un *trigger* com pot ser un *hover, active, target, focus,...*

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https://developer.mozilla.org/es/docs/Web/CSS/CSS_Transitions/Using_CSS_transitions

https://www.w3schools.com/css/css3_transitions.asp

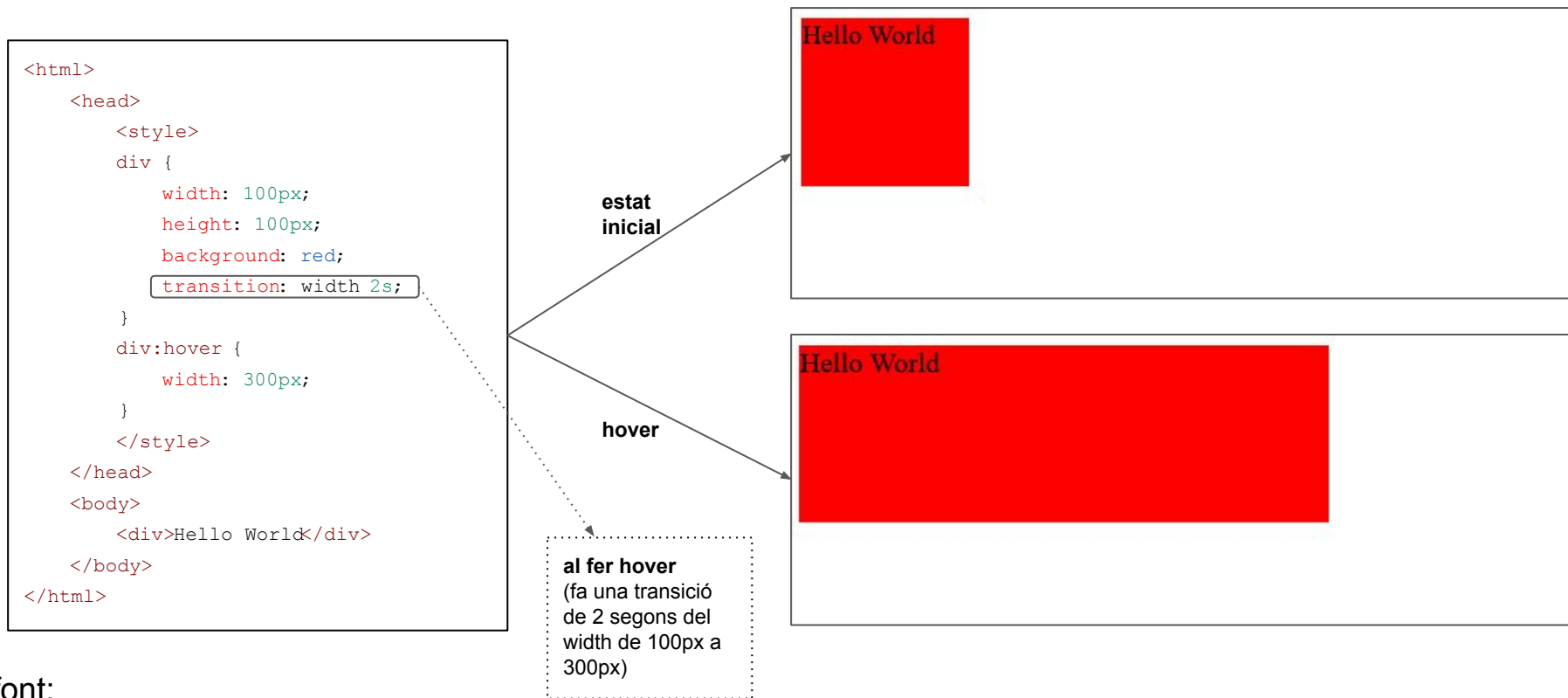
Les propietats principals son:

- transition (*property duration timing-function delay|initial|inherit*)
- transition-delay
- transition-duration
- transition-property
- transition-timing-function

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https://developer.mozilla.org/es/docs/Web/CSS/CSS_Transitions/Using_CSS_transitions

https://www.w3schools.com/css/css3_transitions.asp



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https://developer.mozilla.org/es/docs/Web/CSS/CSS_Transitions/Using_CSS_transitions

https://www.w3schools.com/css/css3_transitions.asp

Les animacions **permeten definir efectes o transicions entre els valors de les propietats** dels estils.

Permeten controlar tot el flux de l'animació, des de que comença fins que acaba.

No necessita un *trigger* per començar l'animació.

font:

https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Animations/Using_CSS_animations

https://www.w3schools.com/css/css3_animations.asp

Les propietats principals son:

- @keyframes
- animation-name
- animation-duration
- animation-delay
- animation-iteration-count
- animation-direction
- animation-timing-function
- animation-fill-mode

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https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Animations/Using_CSS_animations

https://www.w3schools.com/css/css3_animations.asp

@keyframes

Quan especifiqueu estils CSS dins de la regla @keyframes, l'animació canviarà gradualment de l'estil actual al nou en un determinats moments.

```
@keyframes example {  
  0%   {background-color: red;}  
  25%  {background-color: yellow;}  
  50%  {background-color: blue;}  
  100% {background-color: green;}  
}  
  
/* The element to apply the animation to */  
div {  
  background-color: red;  
  animation-name: example;  
  animation-duration: 4s;  
}
```

font:

https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Animations/Using_CSS_animations

https://www.w3schools.com/css/css3_animations.asp

Important

Per a que les animacions no tornin a les propietats inicials hem de fer servir:

```
animation-fill-mode: forwards;
```

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https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Animations/Using_CSS_animations

https://www.w3schools.com/css/css3_animations.asp

HTML5

```
<video>  
  <source src="movie.mp4" type="video/mp4">  
  <source src="movie.ogv" type="video/ogg">  
  Your browser does not support the video tag.  
</video>
```

FORMAT	Media Type
MP4	video/mp4
WebM	video/webm
Ogg	video/ogg

```
<body>
  <button onclick="playVid()" type="button">Play Video</button>
  <button onclick="pauseVid()" type="button">Pause Video</button>
```

```
<video id="myVideo" width="320" height="176">
  <source src="mov_bbb.mp4" type="video/mp4">
  <source src="mov_bbb.ogg" type="video/ogg">
  Your browser does not support HTML5 video.
</video>
</body>
```

```
<script>
var vid = document.getElementById("myVideo");

function playVid() {
  vid.play(); ←
}

function pauseVid() {
  vid.pause(); ←
}
</script>
```